

AP Statistics (Virtual)

2026–27 · The Cushman School

TEACHER

Mr. Willie Avendano

EMAIL

wavendano@cushmanschool.org

CLASS MEETS

8:30–9:30 · Monday asynchronous · Wednesday & Friday live

EXTRA HELP

3:15–4:00 PM on Mondays, Tuesdays, and Wednesdays, or by appointment. *Dismissal runs 3:00–3:15, so help begins at 3:15. Thursdays are unavailable — HS faculty meeting.*

Course Description

The virtual section of AP Statistics. The curriculum, pacing, materials, and AP Exam are identical to the in-person section — the delivery is what differs: three contact hours a week, as one asynchronous Monday plus two live sessions on Wednesday and Friday.

AP Statistics follows the College Board's **Fall 2026 Course and Exam Description**. Students learn to collect, analyze, and draw conclusions from data across four big ideas: exploring data, sampling and experimentation, probability and simulation, and statistical inference. The AP Exam is fully digital, taken in the Bluebook application, on **Tuesday, May 11, 2027 at 12:00 PM**.

Learning Objectives

By the end of this course, students will be able to:

1. Explore and analyze data
2. Sample and experiment
3. Anticipate patterns using probability and simulation
4. Make statistical inferences
5. Use statistical tools
6. Apply statistical concepts to real-world scenarios
7. Communicate statistical findings
8. Think critically about data
9. Choose appropriate statistical methods

Units

Unit 1 Exploring One-Variable Data and Collecting Data

- Investigative questions and variables
- Graphs for categorical and quantitative data
- Describing and comparing distributions
- Random sampling and experimental design

Unit 2 Probability, Random Variables, and Probability Distributions

- Two categorical variables and simulation
- Probability, conditional probability, and independence
- Random variables
- The binomial and normal distributions; the CLT

Unit 3 Inference for Categorical Data: Proportions

- Estimators and sampling distributions
- Confidence intervals for a proportion
- Significance tests and p-values; errors
- Two proportions; chi-square tests

Unit 5 Regression Analysis (taught before Unit 4)

- Scatterplots, correlation, and regression
- Residuals and least squares

Unit 4 Inference for Quantitative Data: Means

- Sampling distributions for means
- Tests for a mean
- Two means: intervals and tests

Week-by-week pacing — including which weeks carry no new content — is on the course page.

Required Texts and Links

Subject to change and grow.

- *The Practice of Statistics*, Seventh Edition (Starnes & Tabor)

- Skew The Script — lessons and YouTube channel
- JMP (statistical software)
- R / RStudio
- AP Classroom (Progress Checks and unit reviews)

Course Materials

- Laptop with a working camera and microphone
- Reliable internet connection
- Graphing calculator (or Desmos)
- Notebook, pencil, pen

Grades and Evaluations

Grades will be calculated using the 4.0 scale as mentioned in the student handbook.

70% Summative unit tests, the December data investigation, the AP-style practice exam, and the post-exam capstone project	25% Formative problem sets, FRQ practice, AP Classroom Progress Checks, participation in live sessions, and unannounced quizzes	5% Diagnostics Entrance and exit tickets, and other checks that are neither summative nor formative.
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Description of Common Assessments

Unit tests are taken in the Friday live session of the week they fall in. The in-person section tests on Thursdays; this section has no Thursday meeting, so every unit test moves to that week's live Friday. Dates are on the course page.

Monday is asynchronous and it still counts. The week's deck, assigned practice, and any recorded walkthroughs are the Monday assignment — the course page lists exactly what to cover. Come to Wednesday's live session with questions from it.

Free-response practice is continuous. AP Statistics is graded on communication as much as computation — a correct number with no justification earns little credit, so we practice writing the reasoning out from day one.

Classwork (CW in Veracross) should be completed during the allotted class time and is **graded for accuracy**. Since these assessments are formative, students will have the chance to resubmit work within an allotted time determined by the teacher.

Homework (HW in Veracross) assignments allow students to practice their skills outside of the school setting or prepare themselves for the following class. Homework assignments are due at the start of the class period.

Scoring on homework will be as follows:

- **4** — All components of the assignment are completed in full, addressing all requirements with a high level of detail.
- **3** — Most components of the assignment are completed, meeting most of the requirements with satisfactory detail.
- **2** — Some components of the assignment are completed, but gaps or missing elements are noticeable. Detail is lacking in some areas.
- **1** — Few components of the assignment are completed, and significant gaps or missing elements are evident. Details are minimal.
- **NTI** — No assignment is turned in and is subject to the late work policy.

Rules and Routines

Your success in this course will directly reflect on the amount of time and energy you put into it.

Rules

Respect. This is the number one rule in the class. Respect you, your peers, and Mr. Avendano.

Consequences. Most violations will undergo the following consequences:

1. **1st Offense:** Verbal warning
2. **2nd Offense:** Conference at the end of class
3. **3rd Offense:** Behavior report to administration

Routines

1. **Attendance in live sessions.** Wednesday and Friday sessions begin at 8:30. Join on time, with your camera on unless we have arranged otherwise, in a space where you can talk through problems.
2. **Asynchronous Mondays.** There is no live meeting. Work through the week's posted materials independently; the course page is the standing reference and works even when Veracross does not.
3. **Communication.** Email is the primary channel. I respond within 24 hours during the school week; if you write on a Friday, expect a reply by Monday.
4. **Your workspace.** Join from a quiet space with your laptop charged and your materials ready. Treat the live session as class: no other tabs, games, or messaging while we are working.
5. **Participation.** There are many ways to participate in class and I expect students to partake in some way. This can include adding to class discussions, asking questions about class materials, working collaboratively during group work, attending extra help, etc.
6. **Absences.** Consistent attendance is strongly recommended as it facilitates the learning process. It is the student's responsibility to inform me via email or in person when they are or will be absent. Students will have two class period days to make up work from absences (Student Handbook, p. 21). Any work handed in afterwards will be subject to the "Late Work Policy."
7. **Late Work.** Students will have one week to submit any late work for partial credit. Late formative assessments can receive up to a 2.0 in the gradebook. It is not fair to give the same credit to students who turn their work in on time as to those who turn their work in late. However, if a student has an *extenuating circumstance*, I am

more than willing to make accommodations. See me BEFORE the due date if this applies to you! **Late work will NOT receive credit after one week.** After the week, the highest possible grade is a 0. In this case, the student will receive a Complete/No Credit or an NTI if never turned in.

8. **Resubmissions/Test Corrections.** If a student obtains a score of 2 or lower on a written exam, they may request to complete test corrections only by filling out the necessary Resubmit Form. These are done only during extra help and in my presence. Students must arrange a time with me to complete corrections within a given deadline and will earn 1/2 credit back on their missed points. The original work must have been submitted on time to resubmit. Students who want to improve an unannounced quiz score may retake a similar quiz (once). This policy only applies to the aforementioned assignments. It does not apply to regular assignments, announced quizzes, or projects.

Academic Integrity

Students are expected and required to be honest in the preparation of assignments, papers, projects, and all other assessments. Cheating (including plagiarism) in any form is not allowed and it hinders the learning process. If a student chooses to do so, the work will receive a 0 and will lead to further disciplinary consequences. If a student cheats off a classmate's work, both will receive the above-mentioned outcomes.

Disclaimer

The institution and teacher reserve the right to make modifications to this information throughout the academic school year, should it be necessary.

AP Student Expectations Agreement

You are currently registered to take AP Statistics (Virtual) this year. Taking an AP course and exam is a collaborative effort between you, the student, your parent/guardian, and the school. Each party plays a role and must commit to meet the expectations noted below.

STUDENT RESPONSIBILITIES

1. **Understanding Course Rigor.** The student acknowledges that AP Statistics (Virtual) is a college-level course that requires significant dedication, time management, and consistent effort. The student understands the rigor involved and agrees to manage their time and effort to successfully complete all course requirements.
2. **Commitment to the AP Exam.** The student agrees to take the AP Statistics (Virtual) exam in May as outlined by the College Board. The expectation is that all enrolled students will participate in the AP exam to measure their understanding and mastery of the course content.
3. **Additional Meeting Times.** The student agrees to attend additional meetings outside of regular school hours as needed. These sessions are designed to supplement learning, provide additional support, and ensure students are adequately prepared for the AP exam.
4. **Coursework and Assignments.** The student will complete all assignments, readings, and projects outside of class time. The student will notify the teacher immediately if they fall behind in class readings and/or assignments.

PARENT/GUARDIAN RESPONSIBILITIES

The Parent/Guardian agrees to be familiar with and accept the AP course requirements and policies, and to help their child organize study time in support of class assignments and additional meeting times outside of the school day.

SCHOOL RESPONSIBILITIES

The School (AP teacher and AP coordinator) agrees to provide rigorous instruction and challenging course content as described in the AP Course Description. The school will provide the student with a copy of the Bulletin for AP Students and Parents and agrees to administer the AP Exam in a fair and secure environment as outlined in the AP Coordinator's Manual. The school will cover the cost of the exam to ensure all students can take the exam without financial burden.

Acknowledgement of Receipt and Understanding

Please sign and submit as soon as possible, but **no later than midnight on Sunday, August 30, 2026**.

Midnight means midnight. Anything submitted at 12:01 AM Monday or later is late. Submissions are checked Monday morning.

I have read and fully understand the academic and behavioral expectations outlined in Mr. Avendano's AP Statistics (Virtual) syllabus. I agree to abide by these guidelines or will undergo the academic and/or disciplinary consequences.

Student Printed Name

Student Signature

Parent/Guardian Printed Name

Parent/Guardian Signature

Date

Teacher Printed Name

Teacher Signature
